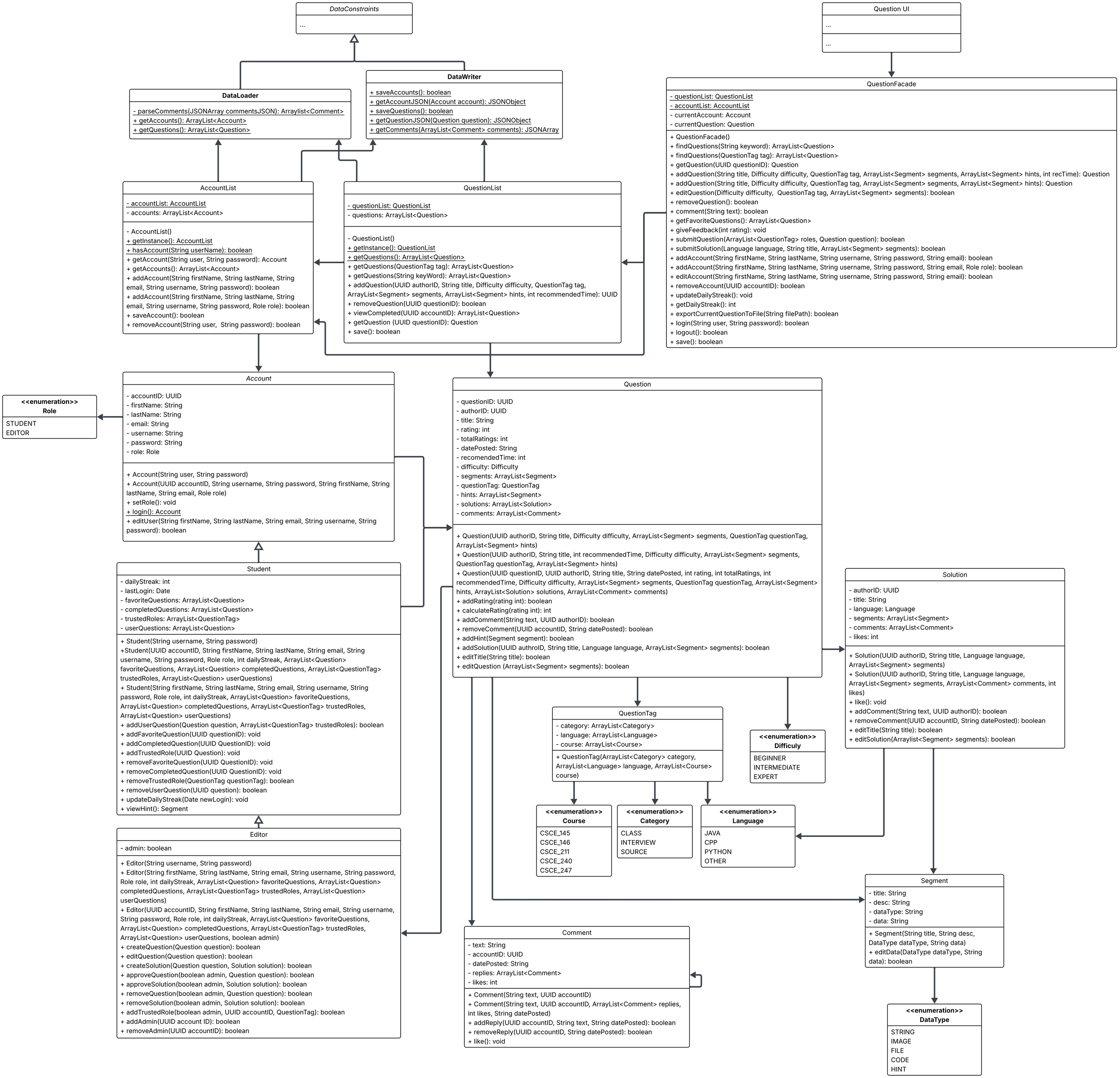
DataConstraints
...
Question UI
...
...
DataLoader
- parseComments(JSONArray commentsJSON): Arraylist<Comment>
+ getAccounts(): ArrayList<Account>
+ getQuestions(): ArrayList<Question>
DataWriter
+ saveAccounts(): boolean
+ getAccountJSON(Account account): JSONObject
+ saveQuestions(): boolean
+ getQuestionJSON(Question question): JSONObject
+ getComments(ArrayList<Comment> comments): JSONArray
QuestionFacade
- questionList: QuestionList
- accountList: AccountList
- currentAccount: Account
- currentQuestion: Question
+ QuestionFacade()
+ findQuestions(String keyword): ArrayList<Question>
+ findQuestions(QuestionTag tag): ArrayList<Question>
+ getQuestion(UUID questionID): Question
+ addQuestion(String title, Difficulty difficulty, QuestionTag tag, ArrayList segments, ArrayList hints, int recTime): Question
+ addQuestion(String title, Difficulty difficulty, QuestionTag tag, ArrayList segments, ArrayList hints): Question
+ editQuestion(Difficulty difficulty, QuestionTag tag, ArrayList segments): boolean
+ removeQuestion(): boolean
+ comment(String text): boolean
+ getFavoriteQuestions(): ArrayList<Question>
+ giveFeedback(int rating): void
+ submitQuestion(ArrayList<QuestionTag> roles, Question question): boolean
+ submitSolution(Language language, String title, ArrayList segments): boolean
+ addAccount(String firstName, String lastName, String username, String password, String email): boolean
+ addAccount(String firstName, String lastName, String username, String password, String email, Role role): boolean
+ editAccount(String firstName, String lastName, String username, String password, String email): boolean
+ removeAccount(UUID accountID): boolean
+ updateDailyStreak(): void
+ getDailyStreak(): int
+ exportCurrentQuestionToFile(String filePath): boolean
+ login(String user, String password): boolean
+ logout(): boolean
+ save(): boolean
AccountList
- accountList: AccountList
- accounts: ArrayList<Account>
- AccountList()
+ getInstance(): AccountList
+ hasAccount(String userName): boolean
+ getAccount(String user, String password): Account
+ getAccounts(): ArrayList<Account>
+ addAccount(String firstName, String lastName, String email, String username, String password): boolean
+ addAccount(String firstName, String lastName, String email, String username, String password, Role role): boolean
+ saveAccount(): boolean
+ removeAccount(String user, String password): boolean
QuestionList
- questionList: QuestionList
- questions: ArrayList<Question>
- QuestionList()
+ getInstance(): QuestionList
+ getQuestions(): ArrayList<Question>
+ getQuestions(QuestionTag tag): ArrayList<Question>
+ getQuestions(String keyWord): ArrayList<Question>
+ addQuestion(UUID authorID, String title, Difficulty difficulty, QuestionTag tag, ArrayList segments, ArrayList hints, int recommendedTime): UUID
+ removeQuestion(UUID questionID): boolean
+ viewCompleted(UUID accountID): ArrayList<Question>
+ getQuestion (UUID questionID): Question
+ save(): boolean
<<enumeration>>
Role
STUDENT
EDITOR
Account
- accountID: UUID
- firstName: String
- lastName: String
- email: String
- username: String
- password: String
- role: Role
+ Account(String user, String password)
+ Account(UUID accountID, String username, String password, String firstName, String lastName, String email, Role role)
+ setRole(): void
+ login(): Account
+ editUser(String firstName, String lastName, String email, String username, String password): boolean
Question
- questionID: UUID
- authorID: UUID
- title: String
- rating: int
- totalRatings: int
- datePosted: String
- recomendedTime: int
- difficulty: Difficulty
- segments: ArrayList
- questionTag: QuestionTag
- hints: ArrayList
- solutions: ArrayList<Solution>
- comments: ArrayList<Comment>
+ Question(UUID authorID, String title, Difficulty difficulty, ArrayList segments, QuestionTag questionTag, ArrayList hints)
+ Question(UUID authorID, String title, int recommendedTime, Difficulty difficulty, ArrayList segments, QuestionTag questionTag, ArrayList hints)
+ Question(UUID questionID, UUID authorID, String title, String datePosted, int rating, int totalRatings, int recommendedTime, Difficulty difficulty, ArrayList segments, QuestionTag questionTag, ArrayList hints, ArrayList<Solution> solutions, ArrayList<Comment> comments)
+ addRating(rating int): boolean
+ calculateRating(rating int): int
+ addComment(String text, UUID authorID): boolean
+ removeComment(UUID accountID, String datePosted): boolean
+ addHint(Segment segment): boolean
+ addSolution(UUID authroID, String title, Language language, ArrayList segments): boolean
+ editTitle(String title): boolean
+ editQuestion (ArrayList segments): boolean
Student
- dailyStreak: int
- lastLogin: Date
- favoriteQuestions: ArrayList<Question>
- completedQuestions: ArrayList<Question>
- trustedRoles: ArrayList<QuestionTag>
- userQuestions: ArrayList<Question>
+ Student(String username, String password)
+Student(UUID accountID, String firstName, String lastName, String email, String username, String password, Role role, int dailyStreak, ArrayList<Question> favoriteQuestions, ArrayList<Question> completedQuestions, ArrayList<QuestionTag> trustedRoles, ArrayList<Question> userQuestions)
+ Student(String firstName, String lastName, String email, String username, String password, Role role, int dailyStreak, ArrayList<Question> favoriteQuestions, ArrayList<Question> completedQuestions, ArrayList<QuestionTag> trustedRoles, ArrayList<Question> userQuestions)
+ addUserQuestion(Question question, ArrayList<QuestionTag> trustedRoles): boolean
+ addFavoriteQuestion(UUID questionID): void
+ addCompletedQuestion(UUID QuestionID): void
+ addTrustedRole(UUID Question): void
+ removeFavoriteQuestion(UUID QuestionID): void
+ removeCompletedQuestion(UUID QuestionID): void
+ removeTrustedRole(QuestionTag questionTag): boolean
+ removeUserQuestion(UUID question): boolean
+ updateDailyStreak(Date newLogin): void
+ viewHint(): Segment
Editor
- admin: boolean
+ Editor(String username, String password)
+ Editor(String firstName, String lastName, String email, String username, String password, Role role, int dailyStreak, ArrayList<Question> favoriteQuestions, ArrayList<Question> completedQuestions, ArrayList<QuestionTag> trustedRoles, ArrayList<Question> userQuestions)
+ Editor(UUID accountID, String firstName, String lastName, String email, String username, String password, Role role, int dailyStreak, ArrayList<Question> favoriteQuestions, ArrayList<Question> completedQuestions, ArrayList<QuestionTag> trustedRoles, ArrayList<Question> userQuestions, boolean admin)
+ createQuestion(Question question): boolean
+ editQuestion(Question question): boolean
+ createSolution(Question question, Solution solution): boolean
+ approveQuestion(boolean admin, Question question): boolean
+ approveSolution(boolean admin, Solution solution): boolean
+ removeQuestion(boolean admin, Question question): boolean
+ removeSolution(boolean admin, Solution solution): boolean
+ addTrustedRole(boolean admin, UUID accountID, QuestionTag): boolean
+ addAdmin(UUID account ID): boolean
+ removeAdmin(UUID accountID): boolean
Solution
- authorID: UUID
- title: String
- language: Language
- segments: ArrayList
- comments: ArrayList<Comment>
- likes: int
+ Solution(UUID authorID, String title, Language language, ArrayList segments)
+ Solution(UUID authorID, String title, Language language, ArrayList segments, ArrayList<Comment> comments, int likes)
+ like(): void
+ addComment(String text, UUID authorID): boolean
+ removeComment(UUID accountID, String datePosted): boolean
+ editTitle(String title): boolean
+ editSolution(Arraylist segments): boolean
QuestionTag
- category: ArrayList<Category>
- language: ArrayList<Language>
- course: ArrayList<Course>
+ QuestionTag(ArrayList<Category> category, ArrayList<Language> language, ArrayList<Course> course)
<<enumeration>>
Difficuly
BEGINNER
INTERMEDIATE
EXPERT
<<enumeration>>
Course
CSCE_145
CSCE_146
CSCE_211
CSCE_240
CSCE_247
<<enumeration>>
Category
CLASS
INTERVIEW
SOURCE
<<enumeration>>
Language
JAVA
CPP
PYTHON
OTHER
Segment
- title: String
- desc: String
- dataType: String
- data: String
+ Segment(String title, String desc, DataType dataType, String data)
+ editData(DataType dataType, String data): boolean
Comment
- text: String
- accountID: UUID
- datePosted: String
- replies: ArrayList<Comment>
- likes: int
+ Comment(String text, UUID accountID)
+ Comment(String text, UUID accountID, ArrayList<Comment> replies, int likes, String datePosted)
+ addReply(UUID accountID, String text, String datePosted): boolean
+ removeReply(UUID accountID, String datePosted): boolean
+ like(): void
<<enumeration>>
DataType
STRING
IMAGE
FILE
CODE
HINT